The Invasion of Snake Robots: Boabot

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BACKGROUND

To go where no other robots can go

Snake robots possess a unique ability no other robot has to slither into any space, climb over large obstacles, and travel uneven terrain. They are a flexible, adaptable class of bio-inspired robots.

Search and rescue

Snake robots have been developed to help in search and rescue sites, but are so far in the early stages of development. Research into search and rescue snake robots is barred by high development costs, so we aim to develop a low cost research platform.

OBJECTIVES

- 💰 Affordable and reproducible
- 🎯 Accurate and reliable odometry
- 🌀 Self-contained and autonomous

IMPACTS AND IMPLICATIONS

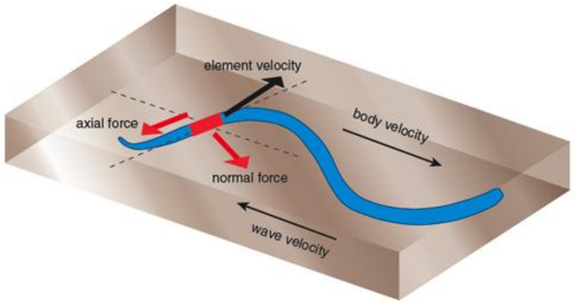
This project shows that a low-cost snake robot can achieve reliable motion, control, and sensing using simple components.

Why it matters: It serves as a scalable platform for applications like search and rescue and inspection.

MOTION

Undulatory Motion

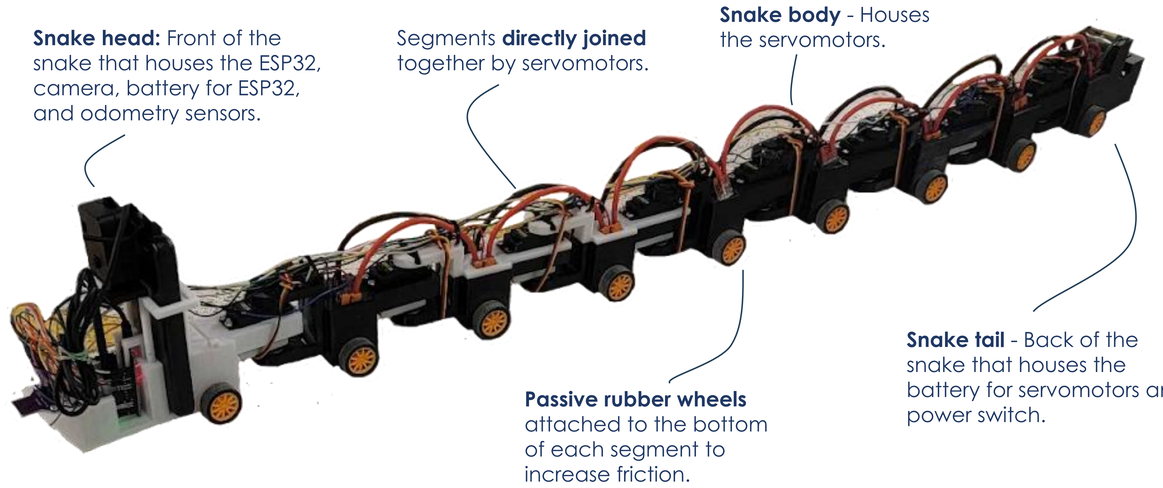
Undulatory Motion uses wave-like movement patterns to propel an animal forward. For our robot, we are using servomotors in our chassis in order to generate these wave-like movement patterns.



MECHANICAL CHASSIS

Design Optimizations

1. **Weight distribution** - The head and tail are by far the heaviest segments and are able to act as **anchor points**, maximizing the **anisotropic force** from the rubber wheels. This prevents the snake head from inadvertently being set off course by any debris on the ground. Meanwhile, the **kinetic force** generated by the actuating body segments contributes to **forward propulsion** rather than lateral slippage.
2. **Fewer but longer segments** - As our robot has fewer but longer segments, there is an **increased power density** and **torque** output at the joints. This further supports the robot's forward propulsion.
3. **Scalability/Modularity - Segments can be easily added or replaced without changing the rest of the robot.** Parts of the head, including the battery brackets, ESP32 bracket, and camera platform, are modular and can be replaced in the case that we want to switch to a different battery, ESP32, or camera.
4. **Separate voltage domains** - The robot consists of a 7.4V voltage domain for the servomotors and a 5V domain for the ESP32 and camera. **This prevents voltage rail instability during full load operation.**



CONTROL

Sensors

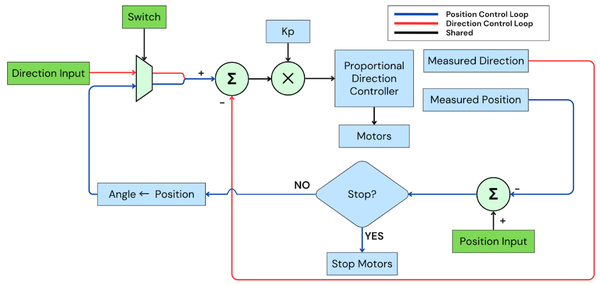
1. **IMU(Inertial Measurement Unit):** This sensor measures the relative **angle change** from its starting location.
2. **Optical Flow Sensor:** This sensor measures the **displacement** of the robot in both x and y direction.

Robot Modes

1. **Manual Control:** The user will use the **joystick** on the **User Interface** to guide the snake to go forward, left, right, and stop.
2. **Direction Control:** The user will give the snake a **direction command of an angle**, which the snake will automatically turn towards and maintain.
3. **Position Control:** The user will give the snake a **(x, y) position pair**, from which the snake will calculate a direction that it must maintain in order to reach the target position. The direction is being computed in real time as the snake moves, making the position navigation robust and reliable.

Proportional Controller

1. In direction control mode, the current measured angle is subtracted from the received target angle, yielding the **angle that the robot needs to turn**. This angle is then multiplied to a constant **Kp**, which is calibrated for smooth turning.
2. The position control operates on the direction control, where the target direction is computed using $atan2((target_y - current_y)(target_x - current_x))$



TESTING & RESULTS

Speed: achieved an average **maximum speed of 34.85cm/s** across 10 trials

Runtime: robot **can operate for up to 36.5 minutes** on a single battery charge (1500mAh)

Hyper-redundancy: **8 DOF** from 8 revolute joints with a maximum rotation range of +/- 90°

Odometry reporting: measuring **accuracy of 99.5% for direction** and **93.5% for distance travelled**

Narrow space navigation: can travel through a **small square hole** with dimensions of **15x15cm**

Vertical mobility: can climb **slopes of 12°**

Camera feed: transmits a camera feed with **24fps at 1280x720p resolution**

NEXT STEPS

Our next steps focus on both **hardware** and **software** improvements.

On the hardware side, we plan to refine the **mechanical design** to enhance robustness and significantly **reduce exposed wiring**, improving both durability and aesthetics. On the software side, we aim to integrate an **LLM-based obstacle avoidance** mechanism to further advance the snake's semi-autonomous capabilities.

In addition, we are considering a redesign that prioritizes **easier accessibility for maintenance** and assembly, while **also reducing overall material costs** to make the system more efficient and scalable.

REFERENCES

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